e-footprint tool pipeline (CO2e) — tools, data, calculation, reporting

Chrome DevTools — manual journey capture

Output: .har (requests, sizes, timings, server IP) + Coverage-*.json (executed JS/CSS code, parsing done later, out of CO2e scope)

Block 1 — Collection tools (collect_env_data.py, sequential)

1. extract_har_data — HAR parsing
2. CrUX (Google API) — device mix
3. ipinfo.io — hosting country/provider
4. detect_tech.py — tech stack (~35 rules)
5. similarweb_api.py (optional) — audience/traffic

Unlike the HAR/Coverage DISPATCH, these tools run one after the other, not in parallel

Block 2 — Collected data -> env-data.json

Page weight, device mix, country/provider/carbon intensity, tech_stack, audience, traffic

Parameters: audience, ios-mobile-share, sw-domain, refresh, check, no-similarweb

Block 3 — run_efootprint.py (e-footprint/Boavizta calculation, reference scenario)

Model (mobile + desktop) -> manufacturing + energy impacts per item (devices, network, server, storage)

Parameters: visits, instance, instance-source, refresh-data

Optional case: third-party generative AI detected -> e-footprint delegates this precise calculation to EcoLogits (third-party genai-impact library), never added to the total without confirmation

Write efootprint-boavizta-model.json + efootprint-python-summary.json

Block 3bis — Recommendation scenarios (recommendations.py, inside run_efootprint.py)

dead_code_reductions, duplicate_reductions, brotli_reductions -> one Reduction per (Job, recommendation) with bytes transferred, bytes decompressed, requests removed

DIRECT decompressed weight (Coverage already measures the dead code on the decompressed) - except HTTP 304 duplicates (cache revalidation), excluded from the removable decompressed weight, counted only in requests and bytes transferred (already tiny)

apply_recommendations -> 3 cumulative SiteSpec (prio1, prio1+2, prio1+2+3), REAL recalculation of the Boavizta model for each

Write efootprint-scenario-{tier}.e-f.json (4 files: reference + 3 tiers, reloadable on e-footprint.boavizta.org) + efootprint-recommendations-gains.json (CO2e gain per tier, reductions_detail per page, excluded recommendations, general hypotheses)

Block 4 — generate_report_html.py (reporting)

Read the CO2e calculation results (efootprint-python-summary.json) and the detected tech stack (tech_stack)

parsing of the Coverage-*.json files for the Dead JS/CSS code section

Read efootprint-recommendations-gains.json (if absent or older schema: sections omitted or "not available", no exception)

Generate the CO2e section (KPIs, manufacturing/energy, hypotheses, tech stack, recommendations)

Recommendations -> block "Estimated CO2e gain" + block "Estimated EcoIndex & LCP gain" (per-page recalculation via har_metrics.py ecoindex_score/ecoindex_grade/lcp_delta_s, LCP = rough estimate, overestimation possible, stated explicitly)

Methodology appendix -> F. Recommendations & CO2e gain, G. EcoIndex & LCP gain per page (3 tables, one per tier, delta per page)